

Program : Diploma in Tool & Die Engineering	
Course Code : 4113	Course Title: Advanced Mould Making
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To introduce the design concepts of injection mould.
- To familiarize students with the concept of Mould Making.
- To impart the aspects of blow moulding and extrusion moulding.
- To introduce various non-conventional injection moulding techniques.

Course Prerequisites:

Topic	Course code	Course name	Semester
Plastic mould		Moulds and dies technology	3
Material		Material science & Metrology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the step-by-step design of various types of injection moulding.	12	Applying
CO2	Demonstrate assembly of various types of injection moulds and process planning.	13	Applying
CO3	Explain different types of blow moulding and extrusion moulding.	20	Understanding
CO4	Explain various types of non-conventional injection moulding techniques.	13	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3		3				
CO2	3		3		3		
CO3	2						
CO4	2						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the step-by-step design of various types of injection moulding.		
M1.01	Illustrate step by step design procedure of injection molds	4	Applying
M1.02	Explain different materials used for inserts	4	Understanding
M1.03	Illustrate the process of Electrode planning	4	Applying
Contents: Step by step design procedure to be followed in the design of injection moulds-shrinkage and its influence in design-feed system design-runner balancing for multicavity and family moulds-gate design-factors to be considered-size, location and number of gating-bolster design-parting surface considerations and venting design-design of cooling system-ejection system design-selection of mould insert materials and mold base materials-electrode planning and factors to be considered in electrode design for sparking.			
CO2	Demonstrate assembly of various types of injection moulds and process planning.		
M2.01	Demonstrate process planning of each individual elements of an injection mould	5	Applying
M2.02	Demonstrate the assembly procedure of various types of injection moulds	5	Applying
M2.03	Explain Molding defects and its remedies	3	Understanding
	Series Test - I	1	
Contents: Process planning and Machining of various components of mould bases-standard components and their types-process charts for each components of moulds for specific components-assembly of two plate, three plate and side core moulds-blue matching-defects and remedies in moulding.			

CO3	Explain different types of blow moulding and extrusion moulding.		
M3.01	Describe stretch and injection orientation blow moulding	6	Understanding
M3.02	Describe miscellaneous blow moulding techniques	6	Understanding
M3.03	Explain the features of advanced extruder machines	5	Understanding
M3.04	Explain the process of co extrusion	3	Understanding

Contents:

Stretch Blow Moulding:

Introduction-Single stage & two stage processes and its comparison-Orientation and stretch ratio-Pre-forming-Extrusion stretch blow molding-Injection orientation blow molding

Co Extrusion blow moulding:

Co-extrusion equipment-Process

Miscellaneous blow moulding processes:

Neck ring process- Drape Process-Dip / displacement processes-Blow molding of irregular shaped parts

Advanced extruded machine features:

Twin screw extruder-Intermeshing and non-Intermeshing-Counter rotating and co-rotating-Comparison with single screw-Vented screw extruder designs-Internal Bubble cooling.

Co Extrusion:

Co-extrusion structures-Barrier materials & adhesives-Comparison: Feed block die and multi manifold die-Advantages of co-extrusion products- Applications of co-extruded products.

CO4	Explain various types of non-conventional injection moulding techniques.		
M4.01	Describe the process of reaction injection moulding	5	Understand
M4.02	Explain the gas assisted injection moulding	4	Understand
M4.03	Explain Multi-colour injection moulding	4	Understand
	Series Test – II	1	

Contents:

Reaction Injection Moulding (RIM):

Introduction to RIM process-Materials and additives

-Features of RIM process and variables-Machine & auxiliary-Flow diagram of RIM process-Characteristic of RIM parts-Merits and demerits of RIM process

Non-conventional injection moulding processes:

Material, process, advantages and disadvantages of the following processes:Water assisted injection moulding- Gas-assisted injection molding-Sandwich injection molding-Structural foam injection molding-Flow molding-Thin wall moulding- Metal filled- Multicolor molding-Injection molding of reinforced thermoplastics.

Text/ Reference:

T/R	Book Title/Author
T1	Injection Mould Design – R.G.W.Pye
R2	How to Make Injection Molds – Georg Menges
R3	Injection Molds : 130 Proven Designs – Hans Gastrow
R4	Runner and Gating Design Handbook – John.P.Beaumont
R5	Injection Moulds for Beginners – Rainer Dangel

Online Resources:

Sl.No	Website Link
1	http://www.mit.edu/
2	https://www.3dhubs.com/guides/injection-molding/